Zachary Scott-Murphy

Computer Visioin Researcher | Athlete

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Education

Georgia Institute of Technology, Computer Science — Atlanta, Georgia, 2025 – 2027

- Simulated strategy game with Agentic Emergent Misalignment LLM.
- Runtime Ceiling of Classical Stereo Correspondence.

University of California, Santa Cruz, Applied Physics — Santa Cruz, California, 2021 – 2024

- Etching technique and Gr/hBN Heterostructure by MMA method.
- Applications of Aligned Bilayer Gr/hBN Devices.

Experience

MPCP, Founding Engineer — UC Berkeley, CA, 2025 – 2026, 1 year

Created the foundation for a live athletic performance system using solely vision. Analysed athletic metric data from world-embodied vision model for the UC Berkeley swimming team.

- UC Berkeley Analytics Conference.
- ACC Swimming Championships 2026.

QuantumGrad, Lead Organiser and Research Communicator — California (Online), 2024 – 2025, 1 year

- Led IBM Qiskit Fall Fest 2024.
- Presented several events with researchers in quantum computing from several universities.
- Wrote articles on quantum computing research and education.
- IBM Quantum Advocate

Awards

College Scholar's Program — Apr 2022

This program gives a select group of students in each year the opportunity to explore research opportunities.
University of California, Santa Cruz
news.ucsc.edu/2021/10/college-scholars-program

Dean's Honors — May 2022

Awarded for top 15% of their academic division.
University of California, Santa Cruz

Publications

Zur Elektrodynamik bewegter Körper

It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.
Albert Einstein
en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Languages

English
Native speaker

Japanese
Proficient

French
Proficient

Interests

Computer Science

Physics

Certificates

MIT iQuHack 2025 Feb 2025

2024 Qiskit Fall Fest Mentor Dec 2024

Qiskit Global Summer School 2024 - Quantum Excellence Aug 2024

Basics of Quantum Information Aug 2024

Quantum Challenge 2024 June 2024

Variational Algorithm Design June 2024

Wastonx.ai Technical Essentials Aug 2024

Projects

MPS/Metal port of Gaussian Splatting (GSplat) from the KAIR Group UC berkeley Apr 2026 – May 2026
Converted CUDA libraries to Metal for all essential libraries. Runtime is comparable to CUDA on an M2 Mac Air 8 gb around 2-3x the runtime of a T4 Tesla GPU.
- Gassuain Splatting
- CUDA
- Metal

Town of Salem Multi-Agent Emergent Misalignment June 2025 – July 2025
Group project in which we created a simulated agent environment mimicking the Town of Salem game, and inserted a variety of different alignments of LLMs, specifically focused on the idea of emergent misalignment models to analyse their competitive strategies and see if they favour chaotic disruptive behaviour as compared to slight-of-hand malicious behaviour to pursue winning.
- LLM Alignment
- Emergent Misalignment
- Multi-Agent Environment

Generative cross-section synthesis for interactive 3D destruction
Generative AI model that took cross-sectional information and surface texture/properties via trilineation to produce fracture patterns for different 3D models and generate the visual colouring and texturing information.
- Generative Modelling
- 3D Modelling
- Ray Tracing

Dark matter literature review
Reviewed all current (as of 2024) materials on dark matter simulation technology to provide a literature review on current strategums and future directions for further research.

- Simulated Physics Computation
- Dark Matter
- Particle Physics

References

Professor Jairo Velasco